

TONKHIL, Mongolia

WMO: 442660

Lat: 46.3128N

Lon: 93.9017E

Elev: 2181

StdP: 77.72

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-27.3	-33.6	0.2	-25.8	-31.8	0.3	-24.9	15.9	-15.0	13.6	-11.2	1.9	0	0.788

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	24.2	12.6	22.8	12.0	21.4	11.5	15.0	21.7	13.8	20.5	12.8	19.3	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	11.6	19.2	10.8	10.6	17.8	9.7	9.8	16.7	50.0	21.9	46.3	20.7	43.0	19.5	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.3	10.6	DB	-31.2	27.4	3.2	1.7	-33.5	28.6	-35.3	29.5	-37.1	30.5	-39.4	31.7	(4)
(5)			WB	-31.3	16.0	2.9	2.1	-33.4	17.5	-35.1	18.7	-36.8	19.9	-38.9	21.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.5	-16.2	-13.1	-5.6	1.9	7.0	13.2	15.6	13.8	8.0	0.6	-6.9	-13.6
	DBStd	11.90	6.16	5.88	5.81	5.37	4.54	3.48	2.86	3.25	4.11	4.67	5.03	5.55
	HDD10.0	3932	812	647	482	244	113	12	1	7	84	291	508	731
	HDD18.3	6535	1070	881	741	493	352	156	92	144	311	549	758	989
	CDD10.0	451	0	0	0	2	19	108	174	124	23	0	0	0
	CDD18.3	12	0	0	0	0	0	2	7	3	0	0	0	0
	CDH23.3	88	0	0	0	0	0	14	53	21	0	0	0	0
	CDH26.7	4	0	0	0	0	0	1	3	1	0	0	0	0
Wind	WSAvg	4.2	3.0	3.7	4.7	5.2	5.5	4.4	4.0	4.2	4.5	4.3	3.9	3.2
Precipitation	PrecAvg	103	1	1	2	2	9	28	29	21	6	1	1	1
	PrecMax	188	4	4	5	8	27	100	81	68	25	6	5	5
	PrecMin	50	0	0	0	0	1	2	1	0	0	0	0	0
	PrecStd	38	1	1	1	2	7	21	19	18	6	1	1	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	2.1	9.8	16.8	21.0	24.9	26.9	25.3	21.1	13.9	8.0	1.2
		MCWB	-4.7	-2.7	2.4	8.0	10.0	13.6	13.8	13.7	9.7	5.7	1.9	-3.1
	2%	DB	-3.2	-0.5	7.4	14.1	18.7	23.2	24.8	23.6	18.6	11.5	4.6	-2.0
		MCWB	-6.6	-4.6	0.6	5.6	8.6	12.1	13.1	11.8	8.0	4.0	-0.1	-5.4
	5%	DB	-5.4	-2.6	5.3	12.1	16.9	21.5	23.2	22.0	16.9	9.8	2.2	-3.9
		MCWB	-8.2	-5.9	-0.7	4.2	7.9	11.6	12.4	11.2	7.7	3.3	-1.9	-7.0
	10%	DB	-7.6	-4.7	3.4	10.2	15.1	19.8	21.7	20.3	15.1	7.9	0.2	-5.8
		MCWB	-9.9	-7.5	-1.7	3.2	6.7	10.9	12.1	10.9	7.1	2.2	-3.5	-8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.1	-2.2	3.6	8.8	11.8	15.6	16.9	16.2	11.2	6.9	2.0	-2.8
		MCDB	-1.4	1.8	9.2	15.6	20.0	23.7	23.4	23.7	18.0	12.4	6.4	0.5
	2%	WB	-6.3	-4.3	0.9	6.5	9.8	14.0	15.5	14.1	10.0	5.2	0.1	-5.2
		MCDB	-3.3	-1.0	6.2	12.9	17.1	21.7	21.9	21.0	16.3	10.3	4.3	-2.2
	5%	WB	-8.1	-5.9	-0.4	4.9	8.5	12.8	14.3	12.8	8.8	3.8	-1.7	-6.9
		MCDB	-5.5	-2.6	5.0	10.7	15.6	19.7	20.6	19.5	14.9	8.8	2.1	-4.2
	10%	WB	-9.9	-7.4	-1.7	3.6	7.3	11.8	13.2	11.7	7.7	2.3	-3.4	-8.3
		MCDB	-7.6	-4.8	3.2	9.4	14.2	17.9	19.5	18.3	13.9	7.5	0.0	-5.9
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.8	11.4	10.6	10.3	9.6	9.2	9.8	10.5	10.3	9.9	9.4
		MCDBR	11.4	12.3	12.7	12.1	12.3	11.9	11.5	12.2	12.6	11.9	11.6	10.8
		MCWBR	9.2	9.7	8.6	7.4	7.1	6.2	5.4	5.8	6.8	7.6	8.5	8.5
	5% WB	MCDBR	11.4	12.0	12.1	11.5	11.6	10.6	9.8	10.6	11.2	11.5	11.2	10.6
		MCWBR	9.3	9.6	8.7	7.9	7.4	6.4	5.9	6.5	6.9	8.1	8.4	8.5
Clear-Sky Solar Irradiance	taub		0.221	0.251	0.290	0.328	0.332	0.341	0.338	0.317	0.286	0.268	0.240	0.224
	taud		2.422	2.353	2.296	2.259	2.311	2.363	2.401	2.441	2.488	2.468	2.450	2.410
	Ebn at Noon		919	941	941	927	932	922	921	930	938	909	880	871
	Edh at Noon		66	88	110	126	124	119	113	104	90	78	64	60
All-Sky Solar Radiation	RadAvg		1.91	3.01	4.47	5.86	6.84	6.91	6.70	6.01	5.08	3.35	2.06	1.54
	RadStd		0.11	0.15	0.23	0.29	0.45	0.37	0.34	0.25	0.20	0.19	0.16	0.08

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	-3.28	N/A	-2.26	-2.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page